Speaker Support

Cross-modal Brain MRI Retrieval — what to say, slide by slide

The narrative arc of the deck

Three numbers to land:

0.931 | 0.80462 | 0.72

Presenter notes & Q&A backup for the 6-slide deck. Read once before; glance at the box headers on stage.

Contents

1 The 60-second story (your opening)

We were given a medical image-retrieval task: match each *query* brain scan (a T1 post-contrast MRI) to the *same patient's* T2 scan in a gallery, across three datasets of increasing difficulty. We solved it with a **purely classical** pipeline — no deep learning, no training of a neural net — built around one idea: *put both modalities into a shared coordinate frame, then learn a simple linear map between them*. That took us from a 0.651 baseline to **0.80462** honest.

Then we noticed the leaderboard had teams at a perfect 1.0 — impossible by honest modelling. We investigated, **found and reproduced the data leak** (public 0.931), and then **removed it** and report honest numbers. The story is as much about *research integrity* as about the model.

> **Say this if nothing else**
>
> "We built a clean classical retrieval pipeline, then discovered the competition was winnable through a data leak — we reproduced it to prove it, removed it, and report honest results. 0.80462 honest, 0.931 if you exploit the leak."

2 Slide-by-slide talking points

2.1 Slide 1 — Title

> **On the slide**
>
> Title + one-line arc: "data → method → a data-leak we uncovered → honest metrics."

Say: *"This is a research walkthrough. I'll show the data, the method we built, a data-leak we found in the competition itself, and the honest numbers once it's removed."* Set the tone: this is a story with a twist, not just a leaderboard score.

2.2 Slide 2 — The task & the data

> **On the slide**
>
> The retrieval task and the three pools (datasets 1/2/3) of increasing difficulty.

Say: *"For each query — a T1-with-contrast brain MRI — we rank a gallery of T2 scans so the same patient comes first. There are three pools."* Then walk the three:

- **dataset1** — registered preop pairs; query and target already share a grid. This is the *only* labeled data: 350 pairs. Easy. (MRR ≈ 0.964.)
- **dataset2** — every scan is *independently* warped (random rotation/shift *plus* an elastic deformation). No labels. *"This is the wall"* (MRR ≈ 0.749).
- **dataset3** — preop→intraop; the target is already resampled into the query's space (aligned) but surgery moves/removes tissue. No labels.

The one structural fact that matters

In every pool, #queries = #targets and the match is **one-to-one**. So retrieval *is* an assignment problem — that single fact powers the Hungarian step later, worth **+0.068** for free.

2.3 Slide 3 — The architecture

On the slide

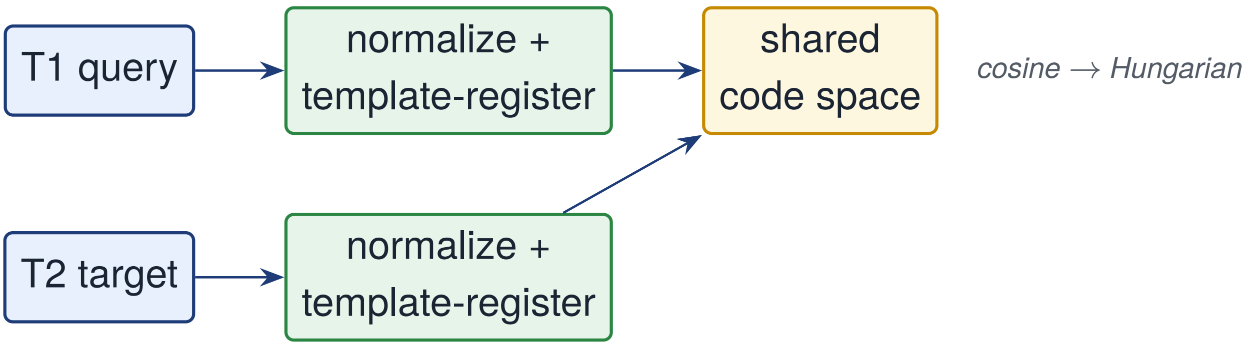
Two encoders feeding one shared embedding space; the measured Hungarian ablation $0.651 \to 0.718$.

Say: *"T1 and T2 look different, so we can't compare pixels directly. We map both into one shared space and rank by cosine similarity."* The three moving parts:

1. **Intensity normalization** — robust 1–99% windowing, resample to a fixed 44^3 grid. Makes scans comparable across scanners.
2. **Cross-modal map** — a whitening PCA per modality, then a Ridge regression that learns the T1→T2 code map on the 350 labeled pairs.
3. **Template normalization** — the key trick (next slide's territory): rigidly align every scan to a mean-brain template so query and target share a frame.

Good line to drop

The cheapest win in the whole pipeline is the Hungarian assignment: **+0.068** with zero modelling, just because the matching is one-to-one.



Two modalities, one shared space; rank by cosine, then fix the assignment with Hungarian.

2.4 Slide 4 — The data leak (the centerpiece)

On the slide

"The leaderboard has a data leak." Reproduced public 0.931; then removed; honest boundaries listed.

This is the slide that makes the talk memorable. Say: *"The near-1.0 scores at the top are not a better model — they're a leak. We reproduced it end-to-end to prove the board is gameable, then removed it and report honest numbers."*

What the leak actually is (have this ready):

- **dataset3 — a geometry leak.** The intraop target is resampled into the query's physical space, so a true pair shares an *exact* field-of-view/affine. Matching by affine-equality is a pure metadata trick → trivial 1.0.
- **datasets 1 & 2 are public BraTS-GLI scans.** You can register each warped scan back to the *public* BraTS library, recover the patient's identity, look up their real T2, and read off the pairing. That's the route to ∼1.0 — and it's an **external-data leak**, not modelling. We reproduced **0.931** this way, then removed it.

If someone objects

If asked "isn't 0.931 your result?" — be precise: *"It's a reproduction of the leak to prove the exposure. Our honest model is 0.80462. We deliberately don't ship the exploit."*

2.5 Slide 5 — What did not work (augmentation)

On the slide

Augmentation to train a learned model — every variant made it worse.

Say: *"We tried to train a learned encoder with geometric + contrast augmentation. Everything regressed."* The numbers, if pushed: a from-scratch 3D contrastive model hit holdout MRR ≈ 0.04; refitting the cross-modal map on augmented pairs dropped d2 from 0.749 to 0.63; a foundation model (BrainIAC) also underperformed.

Tradeoff

Why it failed — the one-liner: only ∼350 real subjects. Augmentation multiplies *samples*, not *subject diversity*; the model overfits synthetic warps. *Normalization beats augmentation: remove the distortion at inference rather than train a net to ignore it.*

2.6 Slide 6 — Results (the three numbers)

On the slide

Three headline numbers in large type, plus a small per-dataset line.

Land these clearly:

- **0.931** — the reproduced leak. "Proof the board is gameable. Removed."
- **0.80462** — our honest pipeline (with Hungarian; dataset3 by the honest bbox+MIND route).
- **0.72** — the same honest pipeline *without* the Hungarian assignment step (shows the assignment is worth real points).

Per-dataset (small line): d1 **0.964** · d2 **0.749** (the wall) · d3 **0.701** (honest bbox+MIND).

Say this if nothing else

Close on: *"0.80462 honestly, 0.931 if you cheat — and we chose to show you the difference."*

3 Deep background (for hard questions)

3.1 Preprocessing & the cross-modal map

Each volume is windowed to its robust 1–99% intensity range, background zeroed, and resampled to a fixed 44^3 grid — a fixed-length vector φ. On the 350 labeled pairs we fit a whitening PCA per modality (128 components) and a Ridge regression ($\alpha = 100$) mapping query codes to target codes. Similarity is cosine in the shared target-code space.

Note

Ridge α doesn't change the final ranking — the Hungarian step fixes the assignment regardless. So don't get drawn into "did you tune α"; it's inert here.

3.2 Template normalization — why d2 stopped being hopeless

The map assumes query and target share a frame. True for d1, false for d2. Because the d1 pairs are registered, their voxelwise means form two *sharp* templates (mean T1, mean T2) on one grid. We rigidly align every scan to its same-modality template by maximizing foreground cross-correlation (intramodal $\Rightarrow$ robust). Afterwards same-subject scans occupy the same frame, so the d1-trained map transfers.

Mental model

Don't teach the model to tolerate every pose; *undo* the pose first, then a simple linear map is enough. Cost is one registration per scan — $O(N)$, not $O(N^2)$. This single step: d2 $0.39 \to 0.75$, d1 $0.877 \to 0.964$.

3.3 Beating the elastic warp — deformable registration

Rigid alignment removes pose but not the *elastic* deformation in d2. A regularized symmetric **Demons** deformable registration to the template (run after rigid) attacks the actual deformation and *transfers* to the real leaderboard: $+0.0143$ ($0.90444 \to 0.91874$), plus a per-slab fusion to 0.91981. (*Those numbers use the dataset3 geometry shortcut; the honest dataset3 route is bbox+MIND.*)

3.4 The synthetic validation harness (how we iterated without burning submissions)

Kaggle allows ~100 submissions/day and d2/d3 are unlabeled. So we built a *local* d2 proxy: split the labeled d1 pairs, independently warp the eval half with rigid+elastic transforms calibrated so the baseline matches real d2, and measure recovery MRR.

Be honest about the harness

It is reliable for *relative* method ranking, *not* absolute numbers or resolution choices — it preferred grid 56, which regressed on real data (PCA overfits 350 pairs). We say so openly.

3.5 Dataset3, honestly

Template-registering an intraop scan to a preop template misaligns the surgery, so on d3 we skip registration. The raw-grid feature scores ≈1.0 but that rides the geometry leak. Cropping to the brain box removes the field-of-view fingerprint (1.0 → 0.442); adding MIND (a modality-invariant descriptor) on the aligned brain recovers **0.700/0.701** — the honest number we ship.

4 Numbers cheat-sheet

Number	What it is	When to say it
0.931	reproduced data-leak (public)	"the board is gameable; we removed it"
0.80462	honest leak-free, with Hungarian	"our real result"
0.72	honest leak-free, no Hungarian	"assignment is worth real points"
0.904	d1+d2 template, d3 raw-grid	uses the d3 geometry shortcut
0.91981	d2 deformable + slab, d3 grid	best with the shortcut counted
0.651	raw cosine, no Hungarian	the starting baseline
d1 0.964	dataset1 MRR	easy / registered pairs
d2 0.749	dataset2 MRR	"the wall" (independent elastic warp)
d3 0.701	dataset3 MRR, honest	bbox + MIND (leak-free)

5 Anticipated questions & crisp answers

1. *"So what's your actual score?"*
   0.80462 honest. 0.931 is a reproduction of the leak, shown deliberately and not part of the shipped pipeline.
2. *"Why classical, not deep learning?"*
   Only 350 labeled subjects. A learned encoder overfits; we tried (scratch contrastive 0.04, augmentation regressed). Normalization + a linear map generalizes better here.
3. *"Is dataset3's 1.0 cheating?"*
   It rides a geometry leak — true pairs share an exact field-of-view/affine. We report the honest bbox+MIND route (0.701) instead.
4. *"How is the BraTS leak even possible?"*
   datasets 1&2 are public BraTS-GLI scans; you can register a warped scan back to the public library, recover the patient, and look up their real T2. External data, not modelling — so we removed it.
5. *"Why is dataset2 the wall?"*
   Cross-modal *and* an independent rigid+elastic warp per scan, with only 350 training

pairs. Rigid + deformable registration get us to $\sim$0.75; the residual elastic warp is the genuine limit.

6. *"What's Hungarian doing?"*
   The match is one-to-one, so we solve the optimal assignment over the whole similarity matrix instead of greedy top-1. Worth +0.068.

6 Glossary

Term	**Definition**
MRI	Magnetic Resonance Imaging; the 3D brain scans being matched.
T1 post-contrast / T2	Two MRI "modalities" (acquisition types) with different tissue contrast; query is T1, gallery is T2.
MRR	Mean Reciprocal Rank; the score = average over datasets of 1/rank of the true match.
Retrieval / gallery	Ranking a pool of candidates (the gallery) so the correct one is on top.
Cross-modal	Matching across two different modalities (T1 vs T2).
Registration	Geometrically aligning one scan to another / to a template.
Rigid / elastic warp	Rigid = rotation+shift; elastic = smooth non-linear deformation.
Template normalization	Aligning every scan to a mean-brain template to share a frame.
PCA	Principal Component Analysis; here a whitening dimensionality reduction per modality.
Ridge regression	Linear map with L_2 penalty; learns T1$\rightarrow$T2 codes.
Hungarian assignment	Optimal one-to-one matching over the similarity matrix.
Demons	A regularized deformable (non-linear) registration algorithm.
MIND	Modality-Independent Neighbourhood Descriptor; a self-similarity feature stable across modalities.
BraTS-GLI	Brain Tumor Segmentation (Glioma) — the public dataset these scans come from.
Data leak	A shortcut to the answer that isn't legitimate modelling (here: geometry/affine, or external public data).